Fair, Accountable, Transparent Machine Learning (FAccT ML)
ZG517

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Session 2
Time – 10:40 AM to 12:40 PM

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

Session Content

- Sensitive Features
- Common Fairness Metrics
- Fairness Thru' Unawareness
  - Difficulty
- Fair Methods for ML
  - Preprocessing Methods
  - In-processing Methods
  - Post-processing Methods

Sensitive (Protected) Features

- Sensitive Features
  - Identify a group
  - e.g., gender, ethnics

- Discrimination Occurs
  - When Sensitive Features Are Used Improperly
  - May lead to ML Discrimination

Protected Attributes Specified in US Fair Lending Laws

- Fair Housing Acts (FHA)
- Equal Credit Opportunity ACts (ECOA)

Attribute	FHA	ECOA
Race	✓	✓
Color	✓	✓
National origin	✓	✓
Religion	✓	✓
Sex	✓	✓
Familial status	✓	
Disability	✓	
Exercised rights under CCPA		✓
Marital status		✓
Recipient of public assistance		✓
Age		✓

Fairness Through Unawareness

- A ML Algorithm Achieves Fair Through Unawareness If
  - None of the sensitive features are directly used in the model

Protected

~~Race and Ethnicity~~	Skills	Years of Exp	Hired?
~~Hispanic~~	Javascript	1	no
~~Hispanic~~	C++	5	yes
~~White~~	Java	2	yes
~~White~~	C++	3	yes

Training

Fair ML Model

Issues With Fairness Through Unawareness

Sensitive Features May Still Be Used

- Inferred from indirect evidence

Inferred

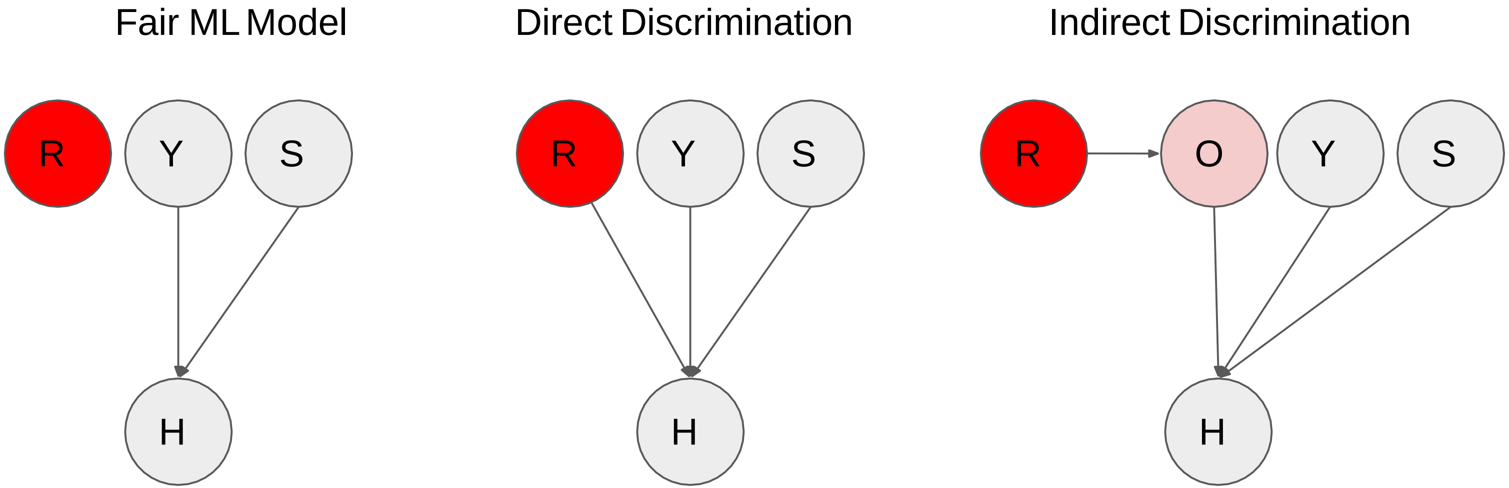
Protected

~~Race and Ethnicity~~	Skills	Years of Exp	Often Goes to Mexican Markets	Hiring Decision
~~Hispanic~~	Javascript	1	yes	no
~~Hispanic~~	C++	5	yes	yes
~~White~~	Java	2	no	yes
~~White~~	C++	3	no	yes

Training

Discriminatory ML Model

Types of Discriminations



R - Race
Y - Years of Exp
S = Skills
O = Often Goes to Mexico Market

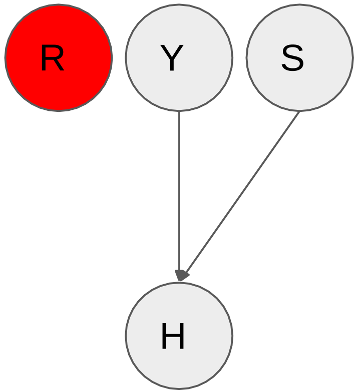
Conditions for Direct Discrimination

- A - set of protected features
- X - set of features other than protected features

- A predictor $\hat{Y}$ is direct discrimination if
  - $P(\hat{Y} \mid X, A) \neq P(\hat{Y} \mid X)$
  - i.e., $\hat{Y} \not\perp\!\!\!\perp A \mid X$

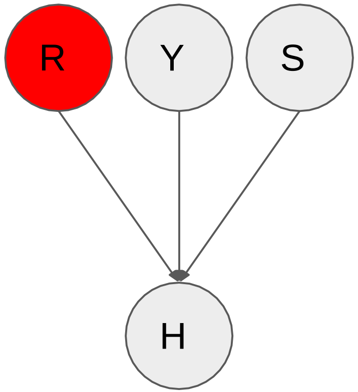
Types of Discriminations

Fair ML Model



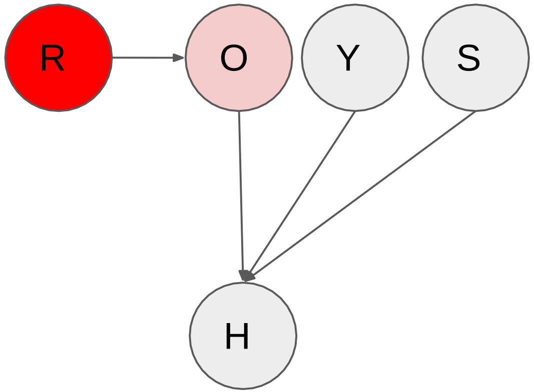
$H \perp\!\!\!\perp R \mid \{Y,S\}$
not connected

Direct Discrimination



$H \not\!\perp\!\!\!\perp R \mid \{Y, S\}$
(head to head)

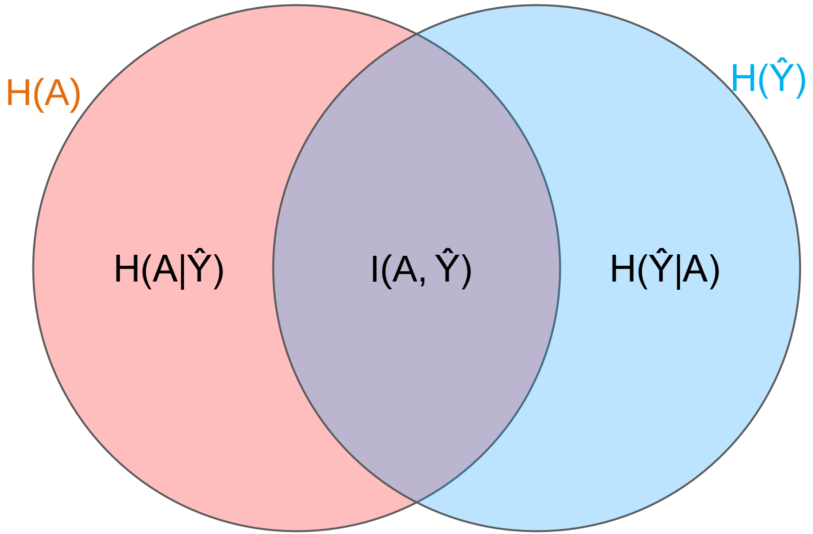
Indirect Discrimination



$H \perp\!\!\!\perp R \mid \{Y,S, O\}$
(head to tail)

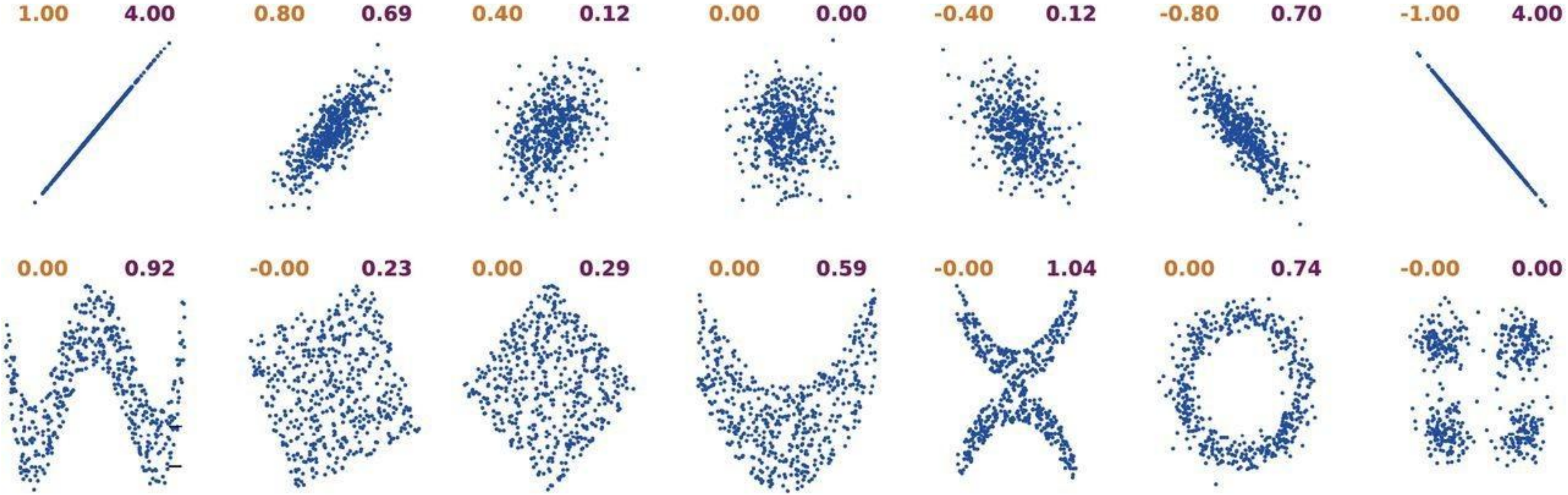
Conditions for Indirect Discrimination

- Mutual Information
  - A measure of the mutual dependence between A and $\hat{Y}$
  - $I(A, \hat{Y}) = H(A) - H(A \mid \hat{Y}) = H(\hat{Y}) - H(\hat{Y} \mid A)$
  - $I(A, \hat{Y}) = 0$ if $P(\hat{Y} \mid A) = P(\hat{Y})$, or $A \perp\!\!\!\perp \hat{Y}$



Correlation Coefficient and Mutual Information

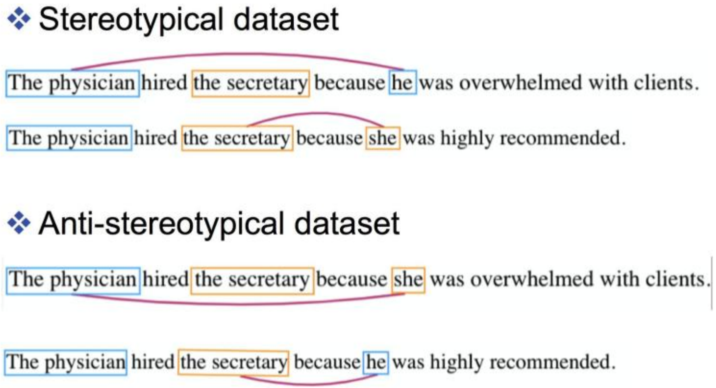
Correlation Coefficient (left) and Mutual Information (right)



Ince et al, 2016

Limitations

- Processing Sensitive Features
  - Fairness through unawareness requires sensitive features to be masked out
  - Not easy to do in real life
  - Referred to as individual fairness criteria



Common Fairness Criteria

- Demographic Parity
- Equality of Odds
- Equality of Opportunity

Demographic Parity Applied to a Group

Demographic Parity Is Applied to a Group of Samples

- Does not require features to be masked out

A Predictor $\hat{Y}$ Satisfies Demographic Parity If

- The probabilities of positive predictions are the same regardless of whether the group is protected
- Protected groups are identified as A = 1

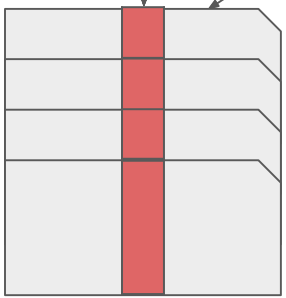
$$P(\hat{Y} = 1 | A = 1) = P(\hat{Y} = 1 | A = 0)$$

Comparisons

Individual Treatment

Protected Features A

Non-protected Features X

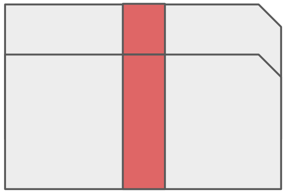


Fairness Through Unawareness

$P(\hat{Y} \mid X)$

Group Treatment

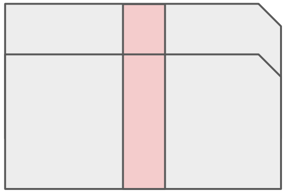
Protected Features = 1



Demographic Parity

$P(\hat{Y}=1 \mid A=1)$

Protected Features = 0



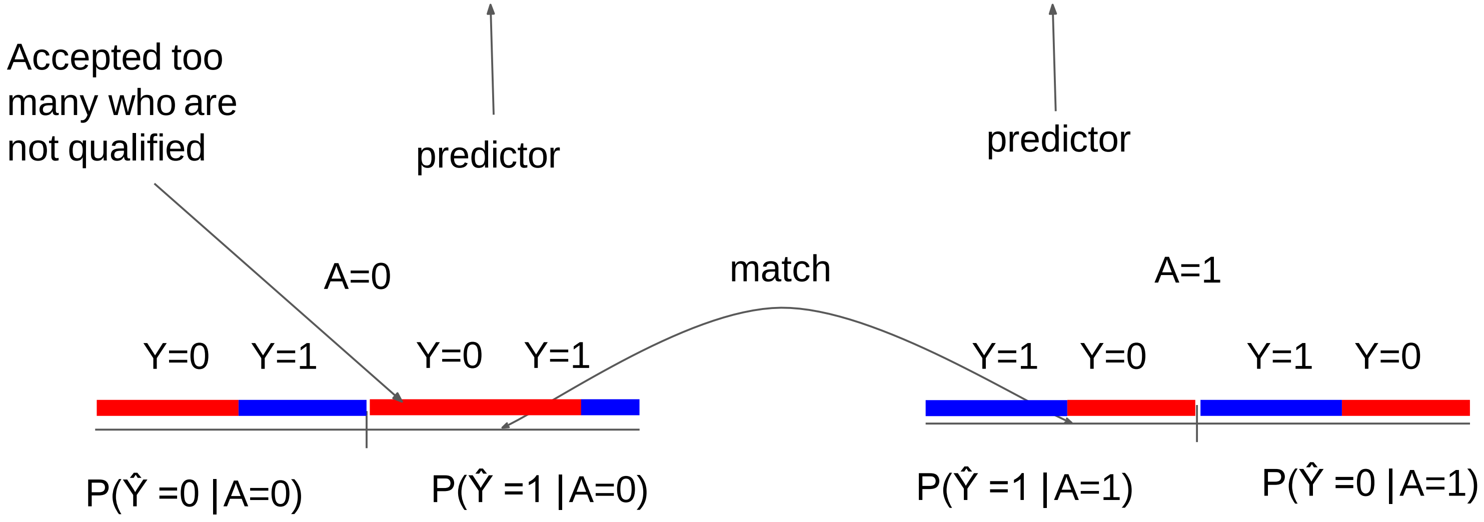
Demographic Parity

$P(\hat{Y}=1 \mid A=0)$

Issues With Demographic Parity

Correlates Too Much With the Performance of the Predictor

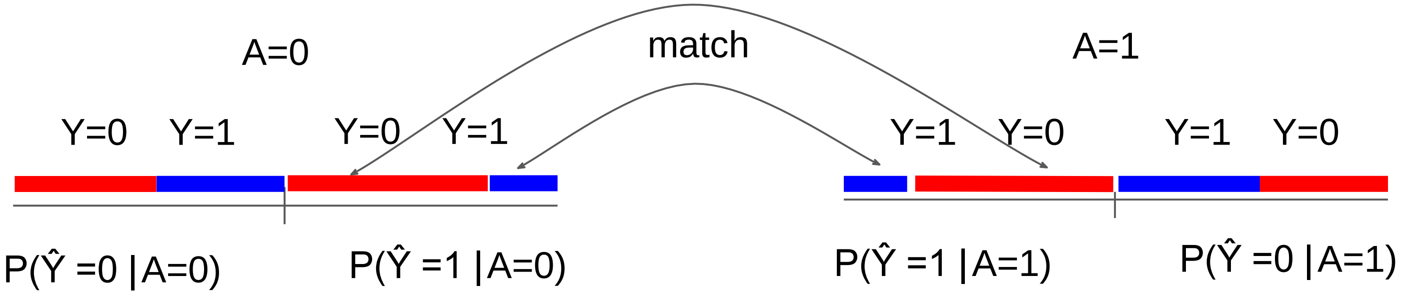
$$P(\hat{Y} = 1|A = 1) = P(\hat{Y} = 1|A = 0)$$



Equality of Odds

Equal Probabilities for Both Qualified/Unqualified People Across Protected Groups

$$P(\hat{Y} = 1 | A = 0, Y) = P(\hat{Y} = 1 | A = 1, Y)$$

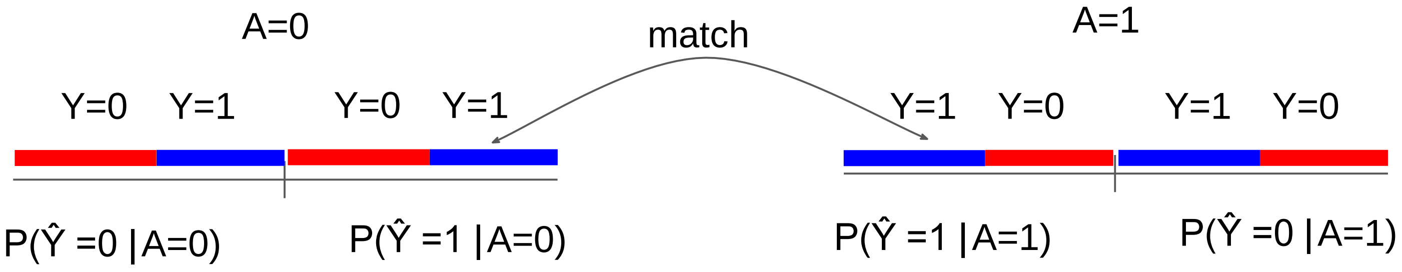


Hardt et al, 2016

Equality of Opportunity

Equal Probabilities for Qualified People Across Protected Groups

$$P(\hat{Y} = 1 | A = 0, Y = 1) = P(\hat{Y} = 1 | A = 1, Y = 1)$$



Hardt et al, 2016

Practice Question

Find out the Fairness Criteria that Ŷ1, and Ŷ2 Satisfy

– A = {race}, Y = {Hiring Decision}

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
Hispanic	Javascript	1	yes	no	0	1
Hispanic	C++	5	yes	yes	1	1
Hispanic	Python	1	no	yes	1	0
White	Java	2	no	yes	0	0
White	C++	3	no	yes	1	1
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Demographic Parity for Predictor Ŷ1

$P(\hat{Y}1 = 1 \mid R = H) = 2/3$

$P(\hat{Y}1 = 1 \mid R = W) =$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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Demographic Parity for Predictor Ŷ1

- $P(\hat{Y}1 = 1 \mid R = H) = 2/3$
- $P(\hat{Y}1 = 1 \mid R = W) = 2/3$

✔ Demographics Parity

$$P(\hat{Y} = 1 | A = 1) = P(\hat{Y} = 1 | A = 0)$$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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Equality of Opportunity/Odds for $\hat{Y}1$

- $P(\hat{Y}1 = 1 \mid R = H, Y = yes) = 1$
- $P(\hat{Y}1 = 1 \mid R = W, Y = yes) =$
- $P(\hat{Y}1 = 1 \mid R = H, Y = no) =$
- $P(\hat{Y}1 = 1 \mid R = W, Y = no) =$

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Equality of Opportunity/Odds for Ŷ1

- $P(\hat{Y}1 = 1 \mid R = H, Y = yes) = 1$
- $P(\hat{Y}1 = 1 \mid R = W, Y = yes) = 0.5$
- $P(\hat{Y}1 = 1 \mid R = H, Y = no) =$
- $P(\hat{Y}1 = 1 \mid R = W, Y = no) =$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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- $P(\hat{Y}1 = 1 \mid R = H, Y = yes) = 1$
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- $P(\hat{Y}1 = 1 \mid R = W, Y = no) = 1$

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Equality of Opportunity/Odds for Ŷ1

- P(Ŷ1 = 1 | R = H, Y = yes) = 1
- P(Ŷ1 = 1 | R = W, Y = yes) = 0.5
- P(Ŷ1 = 1 | R = H, Y = no) = 0
- P(Ŷ1 = 1 | R = W, Y = no) = 1

X Equality of Opportunity

$$P(\hat{Y}=1|A=0,Y=1)=P(\hat{Y}=1|A=1,Y=1)$$

X Equality of Odds

$$P(\hat{Y}=1|A=0,Y)=P(\hat{Y}=1|A=1,Y)$$

Race and Ethnicity	Skill	Years of Exp	Goes to Mexican Markets?	Hiring Decision Y	Predictor $\hat{Y}_1$	Predictor $\hat{Y}_2$
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Summary of Fairness Criteria

Fairness Criteria	Criteria	Group	Individual
Unawareness	Excludes A in Predictions		✓
Demographic Parity	$P(\hat{Y}=1 \mid A=0) = P(\hat{Y}=1 \mid A=1)$	✓	
Equalized Odds	$P(\hat{Y}=1 \mid A=0, Y) = P(\hat{Y}=1 \mid A=1, Y)$	✓	
Equalized Opportunity	$P(\hat{Y}=1 \mid A=0, Y=1) = P(\hat{Y}=1 \mid A=1, Y=1)$	✓	

Required Reading

- Barocas: Ch 2
- Bishop: Ch 8.2
- https://www.borealisai.com/research-blogs/tutorial1-bias-and-fairness-ai/

Recommended

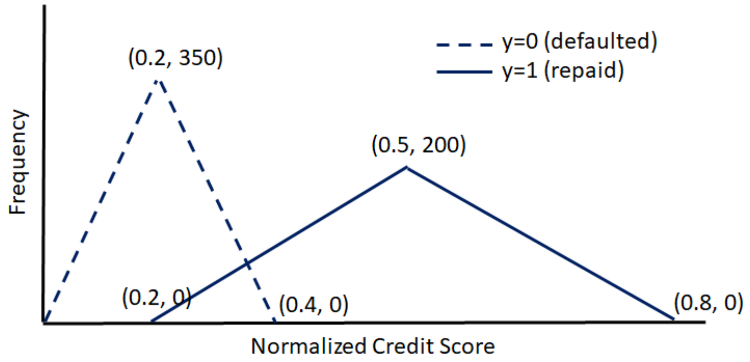
- Papers mentioned on the slides

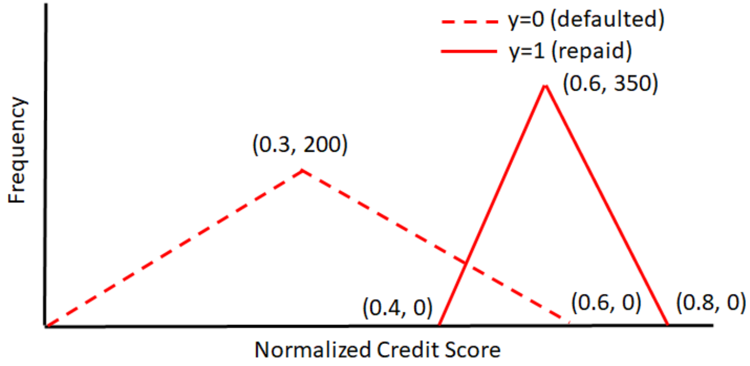
Bishop, Christopher M. Pattern recognition and machine learning. springer, 2006.

Barocas, S., Hardt, M., & Narayanan, A. Fairness and Machine Learning, 2018.

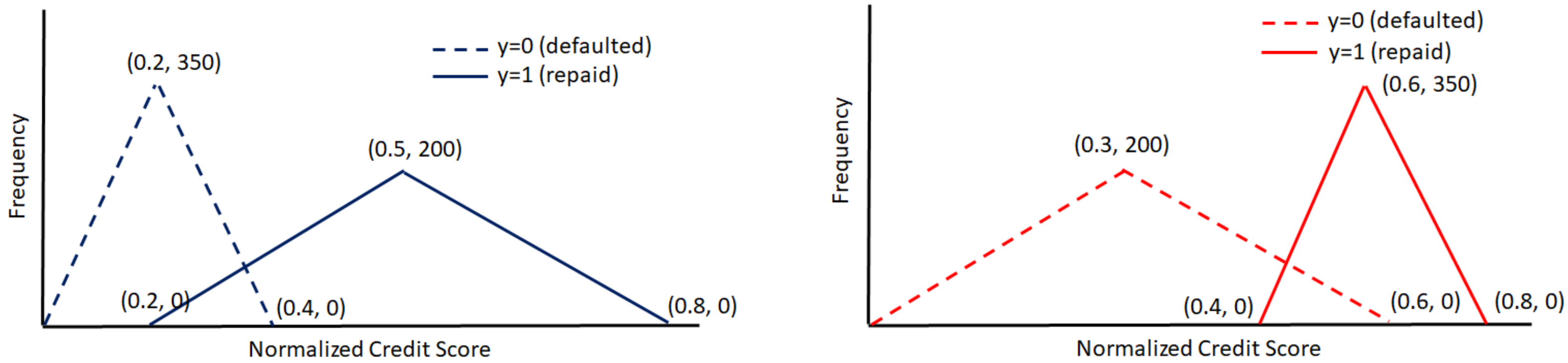
ML EC-3M Problem

Consider a financial institution that uses normalized credit score for approving or rejecting housing loan. Note there are two population of applicants 'blue' (deprived group with protected attribute A=0) and 'red' (favored group with A=1). Loan is approved, i.e., y'=1 if normalized credit score is greater than some threshold t, else rejected. y=1 indicates approved loans that are repaid based on historical data, and y=0 indicates approved loan was defaulted. The financial institution wants to approve loans that is likely to be repaid. Distributions are described in following figure (not drawn to scale) for both populations.





A. Calculate probability P(y'=1) for both 'blue' and 'red' populations for threshold t=0.4. Is fairness achieved w.r.t. protected attribute A?



For 'blue' population,

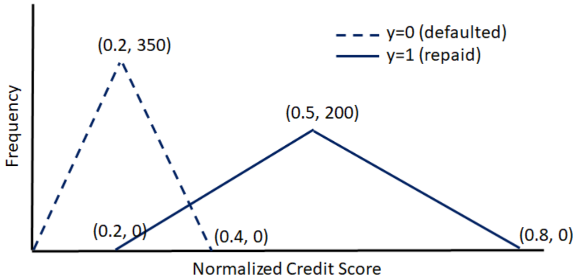
P(y'=1)=[0.5*200*0.6-0.5*0.2*133.3]/[0.2*350+0.3*200]= 0.359

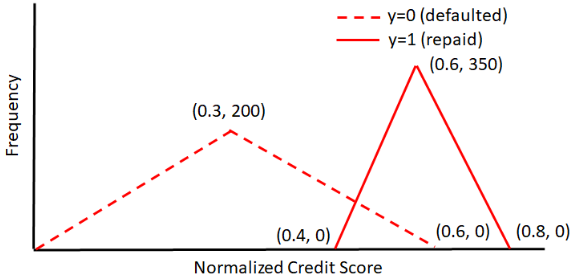
For 'red' population,

P(y'=1)=[0.2*350+0.5*0.2*133.3]/[0.3*200+0.2*350]= 0.641 ➔ Fairness not achieved.

ML EC-3M Problem

B. If threshold t=0.45 is chosen for the blue population, calculate the probability P(y'=1) for the blue population. What threshold value for the red population will ensure demographic parity?





For 'blue' population, for t=0.45

P(Y'=1)=[0.5*200*0.6-0.5*0.25*166.67]/[0.2*350+0.3*200]= 0.30

For 'red population, for any 0.4<t<0.6

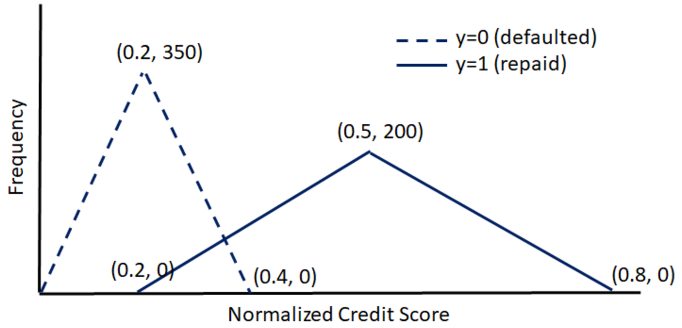
P(Y'=1)=[0.2*350+0.5*(0.6-t)*200/0.3*(0.6-t)-0.5*(t-0.4)*350/0.2]/[0.3*200+0.2* 350]

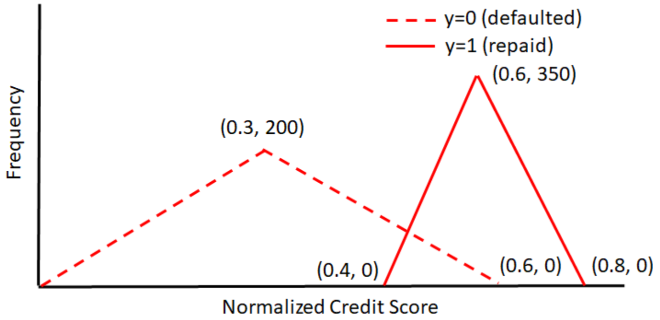
Therefore, for demographic parity,

$70+0.5*(0.6-t)^2*333.33-0.5*(t-0.4)*350/0.2=39$ ➔ t=0.44463

ML EC-3M Problem

C. If threshold t=0.5 is chosen for the blue population, calculate the probability P(y'=1|y=1). What threshold value for the red population will ensure equal opportunity?





For the blue population, from the frequency distribution

- ✓ for t=0.5 P(y'=1|y=1)=0.5

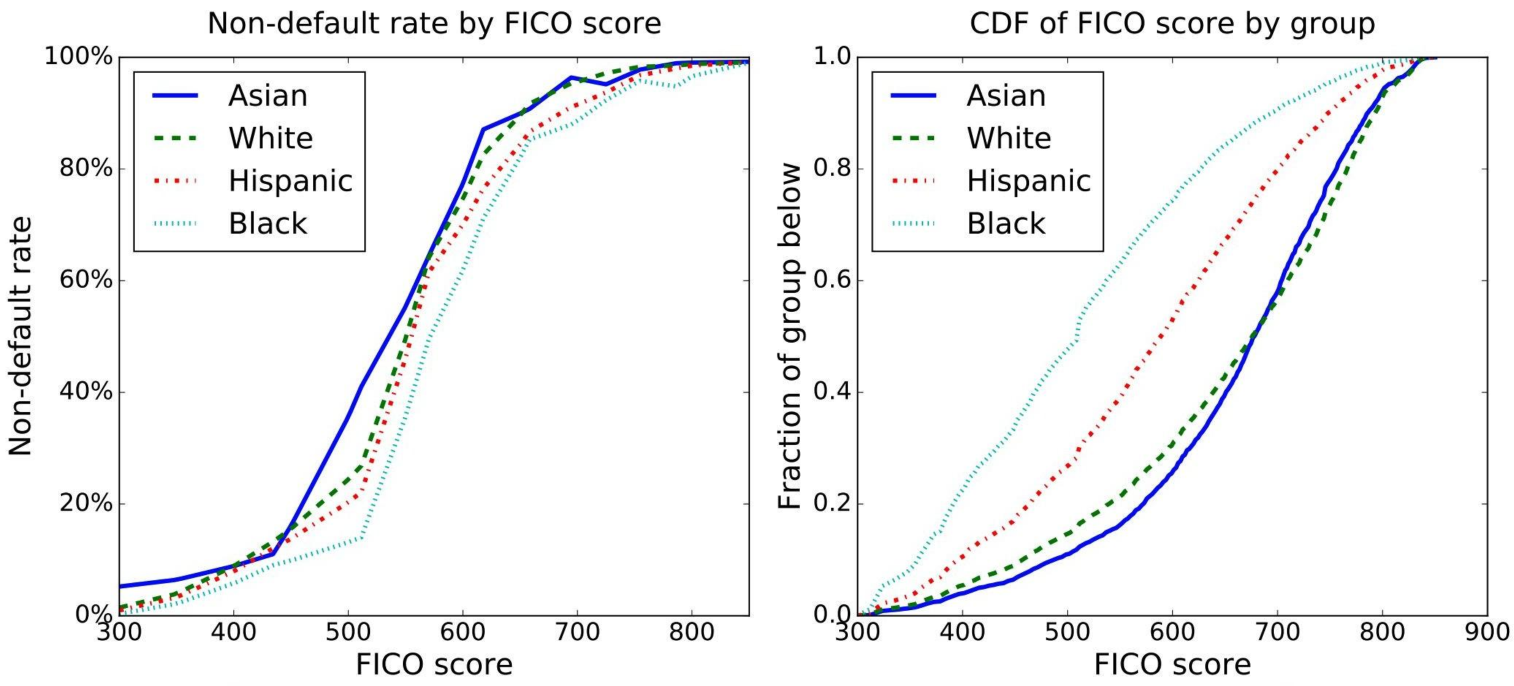
For the red population, from the distribution figure, for t=0.6 P(y'=1|y=1)=0.5

Case Study on FICO

- FICO Dataset
    - 301,536 TransUnion TransRisk scores from 2003
    - Scores ranges from 300 to 850
    - People were labeled as in default if they failed to pay a debt for at least 90 days
    - Protected attribute A is race, with four values: {Asian, white non-Hispanic, Hispanic, and black}

FICO Scores

- 18% Default Rate on Any Accounts Corresponds to a 2% Default Rate for New Loans



Making Lending Decisions Without Discriminating

- Requirement: **Default Rate < 18%**, Simple Threshold Model
  - Max Profit - No Fairness Constraints
  - Race Blind - Using the same threshold for all race groups

Making Lending Decisions Without Discriminating

- Requirement: **Default Rate < 18%**, Simple Threshold Model
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  - Demographic Parity
    - Fraction of the group members that qualify for the loan are the same

$$P(\hat{Y} = 1 | A = 1) = P(\hat{Y} = 1 | A = 0)$$

Making Lending Decisions Without Discriminating

- Requirement: **Default Rate < 18%, Simple Threshold Model**
  - Max Profit - No Fairness Constraints
  - Race Blind - Using the same threshold for all race groups
  - Demographic Parity
    - Fraction of the group members that qualify for the loan are the same

$$P(\hat{Y}=1|A=1)=P(\hat{Y}=1|A=0)$$

  - Equal Opportunity
    - Fraction of non-defaulting group members that qualify for the loan is the same

$$P(\hat{Y}=1|A=0,Y=1)=P(\hat{Y}=1|A=1,Y=1)$$

Making Lending Decisions Without Discriminating

- Requirement: **Default Rate < 18%**, Simple Threshold Model
  - Max Profit - No Fairness Constraints
  - Race Blind - Using the same threshold for all race groups
  - Demographic Parity
    - Fraction of the group members that qualify for the loan are the same

$$P(\hat{Y}=1|A=1) = P(\hat{Y}=1|A=0)$$

  - Equal Opportunity
    - Fraction of non-defaulting group members that qualify for the loan is the same

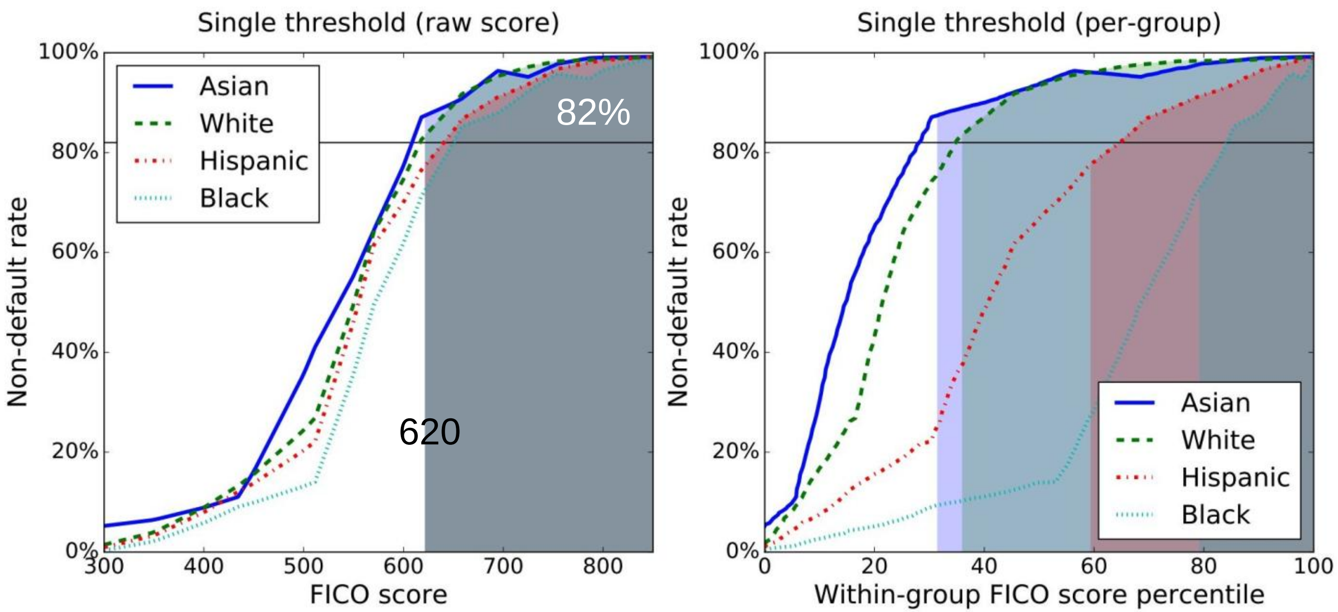
$$P(\hat{Y}=1|A=0,Y=1) = P(\hat{Y}=1|A=1,Y=1)$$

  - Equal Odds
    - Fraction of both non-defaulting and defaulting groups of members that quality for the loan is the same

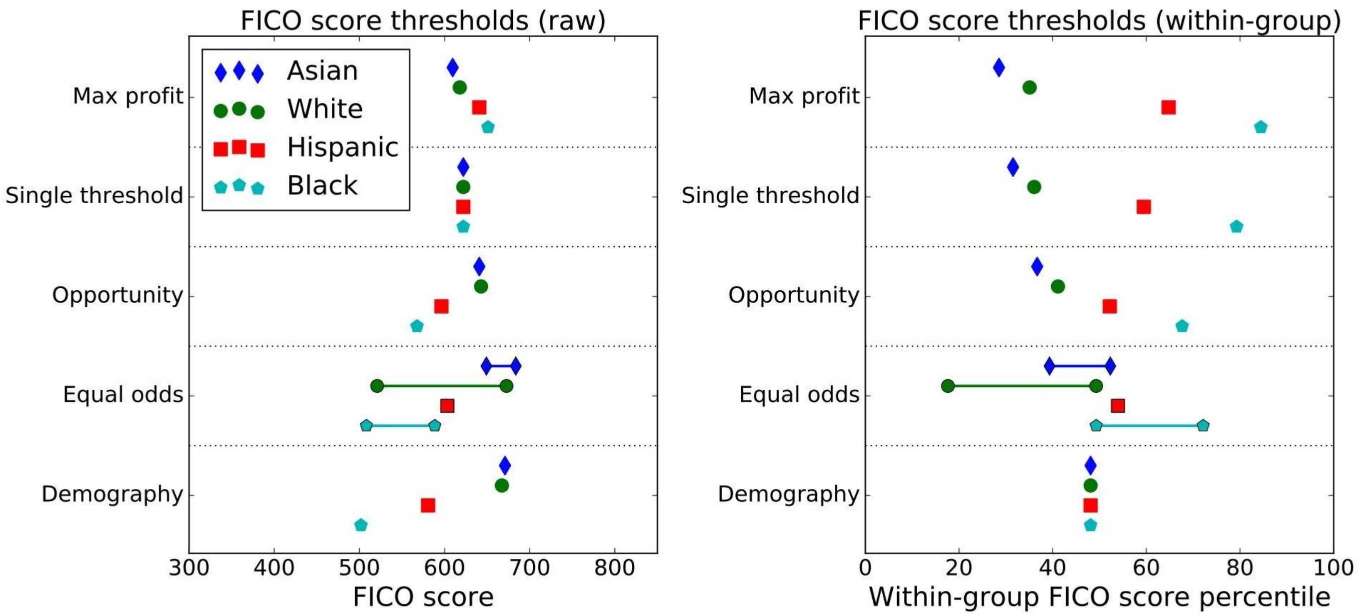
$$P(\hat{Y}=1|A=0,Y) = P(\hat{Y}=1|A=1,Y)$$

Credit Modeling Using A Single Threshold

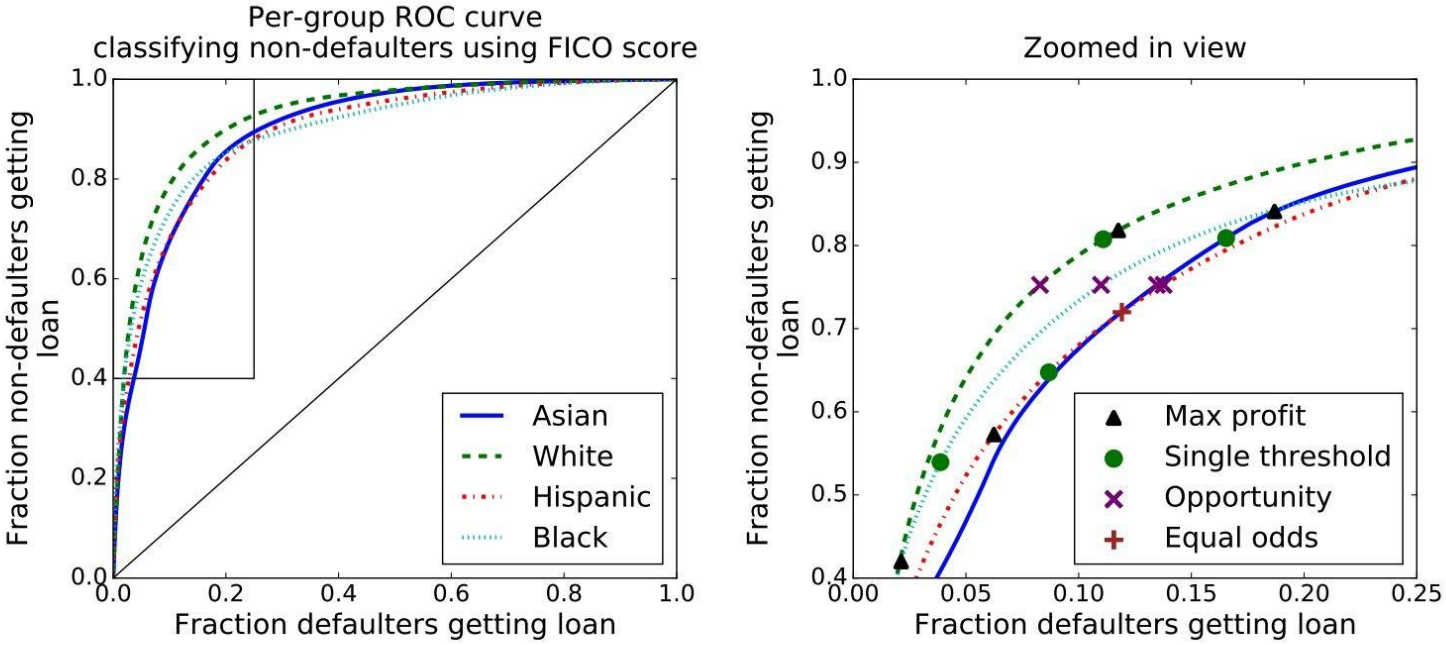
- Within-Group Percentile Differs Dramatically for Each Group



Found Thresholds for Each Fairness Definitions



Identifying Non-Defaulters



Non-Defaulters and Max Profits